

```
CREATE DATABASE ASSESSMENT;
```

```
USE ASSESSMENT;
```

```
CREATE TABLE STUDENT(
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Department VARCHAR(50),
    CGPA DECIMAL(3,2),
    Admission_Year INT
);
```

```
CREATE TABLE COURSE(
    Course_ID VARCHAR(10) PRIMARY KEY,
    Course_Name VARCHAR(100),
    Credit INT,
    Instructor VARCHAR(100),
    Department VARCHAR(50)
);
```

```
INSERT INTO STUDENT (Student_ID, Name, Department, CGPA, Admission_Year) VALUES
(101, 'Rafiq', 'CSE', 3.75, 2021),
(102, 'Nabila', 'EEE', 3.60, 2020),
(103, 'Hasan', 'CSE', 3.90, 2019),
(104, 'Tania', 'BBA', 3.40, 2022),
(105, 'Reza', 'EEE', 3.10, 2021);
```

```
INSERT INTO COURSE (Course_ID, Course_Name, Credit, Instructor, Department) VALUES
('C101', 'Database Systems', 3, 'Dr. Karim', 'CSE'),
('C102', 'Circuit Theory', 3, 'Dr. Rahman', 'EEE'),
('C103', 'Marketing Basics', 2, 'Dr. Jahan', 'BBA'),
('C104', 'Computer Networks', 3, 'Dr. Alam', 'CSE'),
('C105', 'Control Systems', 3, 'Dr. Rahman', 'EEE');
```

```
--1
SELECT
    AVG(CGPA) AS AVG_CGPA
FROM STUDENT
GROUP BY Department;
```

	AVG_CGPA
1	3.400000
2	3.825000
3	3.350000

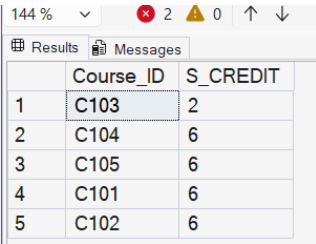
```
--2
SELECT
    Department
FROM STUDENT
GROUP BY Department;
```

	Department
1	BBA
2	CSE
3	EEE

```
--3
SELECT
    S.Student_ID, S.Name, S.Department
FROM STUDENT S
RIGHT OUTER JOIN COURSE C
ON S.Department = C.Department
ORDER BY S.Student_ID ASC;
```

	Student_ID	Name	Department
1	101	Rafiq	CSE
2	101	Rafiq	CSE
3	102	Nabila	EEE
4	102	Nabila	EEE
5	103	Hasan	CSE
6	103	Hasan	CSE
7	104	Tania	BBA
8	105	Reza	EEE
9	105	Reza	EEE

```
--4
SELECT
    c.Course_ID,
    SUM(c.Credit) AS S_CREDIT
FROM COURSE c
LEFT JOIN STUDENT s
    ON s.Department = c.Department
GROUP BY c.Course_ID
ORDER BY S_CREDIT;
```



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	Course_ID	S_CREDIT
1	C103	2
2	C104	6
3	C105	6
4	C101	6
5	C102	6

```
--5
SELECT
    DEPARTMENT,
    MAX(Credit) AS H_CREDIT
FROM COURSE

SELECT * FROM STUDENT;
SELECT * FROM COURSE;
```